TCA9546A Low Voltage 4-Channel I²C and SMBus Switch with Reset Function

1 Features

- 1-of-4 Bidirectional translating switches
- I²C Bus and SMBus compatible
- Active-low reset input
- Three address pins, allowing up to eight TCA9546A devices on the I²C bus
- Channel selection via I²C Bus, in any combination
- Power-up with all switch channels deselected
- Low R_{ON} switches
- Allows voltage-level translation between 1.8-V, 2.5-V, 3.3-V, and 5-V buses
- No glitch on power-up
- Supports hot insertion
- Low standby current
- Operating power-supply voltage range of 1.65 V to 5.5 V
- 5.5 V Tolerant inputs
- 0 to 400-kHz Clock frequency
- Latch-up performance exceeds 100 mA Per JESD 78
- ESD Protection exceeds JESD 22
  - 4000-V Human-body model (A114-A)
  - 1500-V Charged-device model (C101)

2 Applications

- Servers
- Routers (telecom switching equipment)
- Factory automation
- Products with I²C slave address conflicts (multiple, identical temp sensors)

3 Description

The TCA9546A is a quad bidirectional translating switch controlled via the I²C bus. The SCL/SDA upstream pair fans out to four downstream pairs, or channels. Any individual SCn/SDn channel or combination of channels can be selected, determined by the contents of the programmable control register.

An active-low reset ($\overline{\text{RESET}}$) input allows the TCA9546A to recover from a situation in which one of the downstream I²C buses is stuck in a low state. Pulling $\overline{\text{RESET}}$ low resets the I²C state machine and causes all the channels to be deselected, as does the internal power-on reset function.

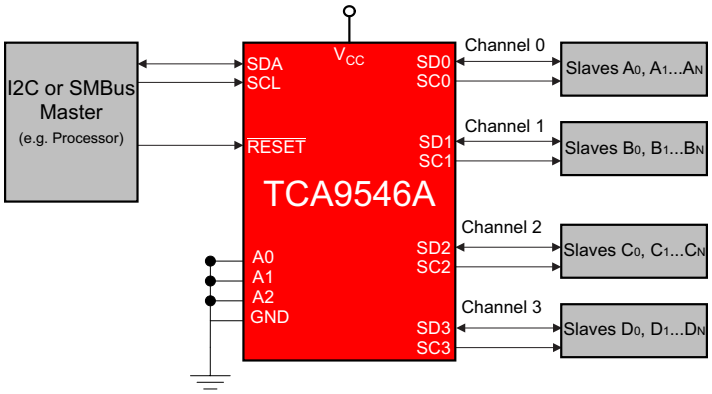
The pass gates of the switches are constructed such that the VCC pin can be used to limit the maximum high voltage, which will be passed by the TCA9546A. This allows the use of different bus voltages on each pair, so that 1.8-V, 2.5-V, or 3.3-V parts can communicate with 5-V parts without any additional protection. External pull-up resistors pull the bus up to the desired voltage level for each channel. All I/O pins are 5.5-V tolerant.

Device Information(1)

DEVICE NAME	PACKAGE	BODY SIZE (NOM)
TCA9546A	TSSOP (16)	5.00 mm x 4.40 mm
	SOIC (16)	9.90 mm x 3.91 mm

(1) For all available packages, see the orderable addendum at the end of the datasheet.

Simplified Application Diagram



An IMPORTANT NOTICE at the end of this data sheet addresses availability, warranty, changes, use in safety-critical applications, intellectual property matters and other important disclaimers. PRODUCTION DATA.

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4 Revision History

Changes from Revision A (February 2015) to Revision B **Page**

- Changed the Pin Configuration image appearance ... 3
- Changed V_{CC} = 3.3 V to V_{CC} = 2.5 V in Figure 15 ... 16

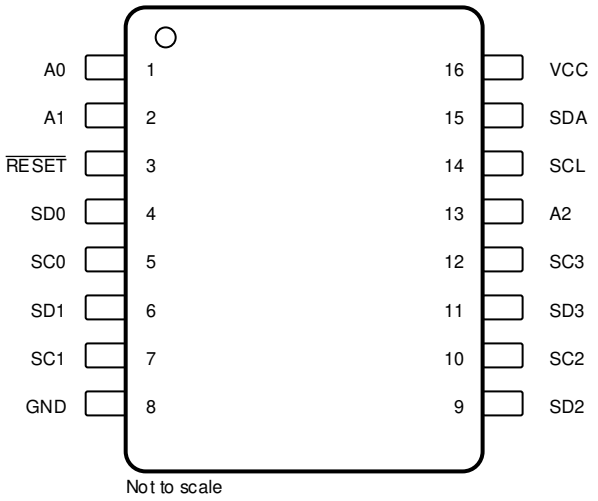
Changes from Original (April 2014) to Revision A **Page**

- Added D package to the datasheet. ... 1
- Changed Handling Ratings table to ESD Ratings. ... 4
- Added D package to the Thermal Information table. ... 4

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5 Pin Configuration and Functions

D or PW Package
TSSOP and SOIC 16 Pins
Top View



Not to scale

Pin Functions

PIN		DESCRIPTION
NAME	NO.	
A0	1	Address input 0. Connect directly to V_{CC} or ground.
A1	2	Address input 1. Connect directly to V_{CC} or ground.
$\overline{\text{RESET}}$	3	Active low reset input. Connect to V_{CC} or V_{DPUM}(1) through a pull-up resistor, if not used.
SD0	4	Serial data 0. Connect to V_{DPU0}(1) through a pull-up resistor.
SC0	5	Serial clock 0. Connect to V_{DPU0}(1) through a pull-up resistor.
SD1	6	Serial data 1. Connect to V_{DPU1}(1) through a pull-up resistor.
SC1	7	Serial clock 1. Connect to V_{DPU1}(1) through a pull-up resistor.
GND	8	Ground
SD2	9	Serial data 2. Connect to V_{DPU2}(1) through a pull-up resistor.
SC2	10	Serial clock 2. Connect to V_{DPU2}(1) through a pull-up resistor.
SD3	11	Serial data 3. Connect to V_{DPU3}(1) through a pull-up resistor.
SC3	12	Serial clock 3. Connect to V_{DPU3}(1) through a pull-up resistor.
A2	13	Address input 2. Connect directly to V_{CC} or ground.
SCL	14	Serial clock line. Connect to V_{DPUM}(1) through a pull-up resistor.
SDA	15	Serial data line. Connect to V_{DPUM}(1) through a pull-up resistor.
VCC	16	Supply power

(1) V_{DPUX} is the pull-up reference voltage for the associated data line. V_{DPUM} is the master I^2C master reference voltage and V_{DPU0}-V_{DPU3} are the slave channel reference voltages.

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6 Specifications

6.1 Absolute Maximum Ratings(1)

over operating free-air temperature range (unless otherwise noted)

		MIN	MAX	UNIT
V_{CC}	Supply voltage range	–0.5	7	V
V_I	Input voltage range(2)	–0.5	7	V
I_I	Input current		±20	mA
I_O	Output current		±25	mA
	Continuous current through V_{CC}		±100	mA
	Continuous current through GND		±100	mA
P_{tot}	Total power dissipation		400	mW
T_A	Operating free-air temperature range	–40	85	°C
T_{stg}	Storage temperature range	-65	150	°C

(1) Stresses beyond those listed under "absolute maximum ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under "recommended operating conditions" is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
(2) The input negative-voltage and output voltage ratings may be exceeded if the input and output current ratings are observed.

6.2 ESD Ratings

			VALUE	UNIT
$V_{(ESD)}$	Electrostatic discharge	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001(1)	±4000	V
		Charged-device model (CDM), per JEDEC specification JESD22-C101(2)	±1500	

(1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process. Manufacturing with less than 500-V HBM is possible with the necessary precautions.
(2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process. Manufacturing with less than 250-V CDM is possible with the necessary precautions.

6.3 Recommended Operating Conditions(1)

			MIN	MAX	UNIT
V_{CC}	Supply voltage		1.65	5.5	V
V_{IH}	High-level input voltage	SCL, SDA	0.7 × V_{CC}	6	V
		A2–A0, $\overline{\text{RESET}}$	0.7 × V_{CC}	V_{CC} + 0.5	
V_{IL}	Low-level input voltage	SCL, SDA	–0.5	0.3 × V_{CC}	V
		A2–A0, $\overline{\text{RESET}}$	–0.5	0.3 × V_{CC}	
T_A	Operating free-air temperature		–40	85	°C

(1) All unused inputs of the device must be held at V_{CC} or GND to ensure proper device operation. Refer to the TI application report, *Implications of Slow or Floating CMOS Inputs*, literature number SCBA004.

6.4 Thermal Information

THERMAL METRIC(1)		TCA9546A		UNIT
		D	PW	
		16 PINS	16 PINS	
$R_{\theta JA}$	Junction-to-ambient thermal resistance	92.3	122.3	°C/W
$R_{\theta JCtop}$	Junction-to-case (top) thermal resistance	52.3	56.6	
$R_{\theta JB}$	Junction-to-board thermal resistance	50.1	57.4	
ψ_{JT}	Junction-to-top characterization parameter	17.7	10.9	
ψ_{JB}	Junction-to-board characterization parameter	49.8	66.8	

(1) For more information about traditional and new thermal metrics, see the *Semiconductor and IC Package Thermal Metrics* application report.

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6.5 Electrical Characteristics(1)

over recommended operating free-air temperature range (unless otherwise noted)

PARAMETER			TEST CONDITIONS		V_{CC}	MIN	TYP(2)	MAX	UNIT
V_{PORR}	Power-on reset voltage, V_{CC} rising		No load,	$V_I = V_{CC}$ or GND(3)			1.2	1.5	V
V_{PORF}	Power-on reset voltage, V_{CC} falling(4)		No load,	$V_I = V_{CC}$ or GND(3)		0.8	1		
V_{pass}	Switch output voltage		$V_{SWin} = V_{CC}$,	I_{SWout} = –100 µA	5 V		3.6		V
					4.5 V to 5.5 V	2.6		4.5	
					3.3 V		1.9		
					3 V to 3.6 V	1.6		2.8	
					2.5 V		1.4		
					2.3 V to 2.7 V	1.0		1.8	
					1.8 V		0.8		
					1.65 V to 1.95 V	0.5		1.1	
I_{OL}	SDA		V_{OL} = 0.4 V		1.65 V to 5.5 V	3	7		mA
			V_{OL} = 0.6 V			6	10		
I_I	SCL, SDA		$V_I = V_{CC}$ or GND(3)		1.65 V to 5.5 V			±1	µA
	SC3–SC0, SD3–SD0							±1	
	A2–A0							±1	
	$\overline{RESET}$							±1	
I_{CC}	Operating mode	f_{SCL} = 400 kHz	$V_I = V_{CC}$ or GND(3) I_O = 0 $t_{r,max}$ = 300 ns		5.5 V		50		µA
					3.6 V		20		
					2.7 V		11		
					1.65 V		6		
		f_{SCL} = 100 kHz	$V_I = V_{CC}$ or GND(3) I_O = 0 $t_{r,max}$ = 1 µs		5.5 V		35		
					3.6 V		14		
					2.7 V		5		
					1.65 V		2		
	Standby mode	Low inputs	V_I = GND(3) I_O = 0		5.5 V		1.6	2	
					3.6 V		1.0	1.3	
					2.7 V		0.7	1.1	
					1.65 V		0.4	0.55	
		High inputs	$V_I = V_{CC}$ I_O = 0		5.5 V		1.6	2	
					3.6 V		1.0	1.3	
					2.7 V		0.7	1.1	
					1.65 V		0.4	0.55	
ΔI_{CC}	Supply-current change	SCL, SDA	SCL or SDA input at 0.6 V, Other inputs at V_{CC} or GND(3)		1.65 V to 5.5 V		2	15	µA
			SCL or SDA input at V_{CC} – 0.6 V, Other inputs at V_{CC} or GND(3)		1.65 V to 5.5 V		2	15	
C_i	A2–A0		$V_I = V_{CC}$ or GND(3)		1.65 V to 5.5 V		4.5	6	pF
	$\overline{RESET}$						4.5	5.5	
$C_{io(OFF)}$ (5)	SCL, SDA		$V_I = V_{CC}$ or GND(3)	Switch OFF	1.65 V to 5.5 V		15	19	pF
	SC3–SC0, SD3–SD0						6	8	

(1) For operation between specified voltage ranges, refer to the worst-case parameter in both applicable ranges.
(2) All typical values are at nominal supply voltage (1.8-V, 2.5-V, 3.3-V, or 5-V V_{CC}), T_A = 25°C.
(3) $\overline{RESET}$ = V_{CC} (held high) when all other input voltages, V_I = GND.
(4) The power-on reset circuit resets the I^2C bus logic with $V_{CC} < V_{PORF}$.
(5) $C_{io(ON)}$ depends on internal capacitance and external capacitance added to the SCn lines when channels(s) are ON.

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Electrical Characteristics(1) (continued)

over recommended operating free-air temperature range (unless otherwise noted)

PARAMETER		TEST CONDITIONS		V_{CC}	MIN	TYP(2)	MAX	UNIT
R_{ON}	Switch on-state resistance	V_O = 0.4 V	I_O = 15 mA	4.5 V to 5.5 V	4	10	16	Ω
				3 V to 3.6 V	5	13	20	
		V_O = 0.4 V	I_O = 10 mA	2.3 V to 2.7 V	7	16	45	
				1.65 V to 1.95 V	10	25	70	

6.6 I²C Interface Timing Requirements

over recommended operating free-air temperature range (unless otherwise noted) (see Figure 5)

			STANDARD MODE I²C BUS		FAST MODE I²C BUS		UNIT
			MIN	MAX	MIN	MAX	
f_{scl}	I²C clock frequency		0	100	0	400	kHz
t_{sch}	I²C clock high time		4		0.6		µs
t_{scl}	I²C clock low time		4.7		1.3		µs
t_{sp}	I²C spike time			50		50	ns
t_{sds}	I²C serial-data setup time		250		100		ns
t_{sdh}	I²C serial-data hold time		0(1)		0(1)		µs
t_{icr}	I²C input rise time			1000	20 + 0.1C_b (2)	300	ns
t_{icf}	I²C input fall time			300	20 + 0.1C_b (2)	300	ns
t_{ocf}	I²C output fall time	10-pF to 400-pF bus		300	20 + 0.1C_b (2)	300	ns
t_{buf}	I²C bus free time between stop and start		4.7		1.3		µs
t_{sts}	I²C start or repeated start condition setup		4.7		0.6		µs
t_{sth}	I²C start or repeated start condition hold		4		0.6		µs
t_{sps}	I²C stop condition setup		4		0.6		µs
$t_{vdL(Data)}$	Valid-data time (high to low)(3)	SCL low to SDA output low valid		1		1	µs
$t_{vdH(Data)}$	Valid-data time (low to high)(3)	SCL low to SDA output high valid		0.6		0.6	µs
$t_{vd(ack)}$	Valid-data time of ACK condition	ACK signal from SCL low to SDA output low		1		1	µs
C_b	I²C bus capacitive load			400		400	pF

(1) A device internally must provide a hold time of at least 300 ns for the SDA signal (referred to the V_{IH} min of the SCL signal), in order to bridge the undefined region of the falling edge of SCL.
(2) C_b = total bus capacitance of one bus line in pF
(3) Data taken using a 1-kΩ pullup resistor and 50-pF load (see Figure 5)

6.7 Switching Characteristics

over recommended operating free-air temperature range, C_L ≤ 100 pF (unless otherwise noted) (see Figure 5)

PARAMETER			FROM (INPUT)	TO (OUTPUT)	MIN	MAX	UNIT
t_{pd} (1)	Propagation delay time	R_{ON} = 20 Ω, C_L = 15 pF	SDA or SCL	SDn or SCn		0.3	ns
		R_{ON} = 20 Ω, C_L = 50 pF				1	

(1) The propagation delay is the calculated RC time constant of the typical ON-state resistance of the switch and the specified load capacitance, when driven by an ideal voltage source (zero output impedance).

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6.8 Interrupt and Reset Timing Requirements

over recommended operating free-air temperature range (unless otherwise noted)

PARAMETER		MIN	MAX	UNIT
t_{WL}	Pulse duration, $\overline{\text{RESET}}$ low	6		ns
t_{rst} (1)	$\overline{\text{RESET}}$ time (SDA clear)		500	ns
$t_{REC(STA)}$	Recovery time from $\overline{\text{RESET}}$ to start	0		ns

(1) t_{rst} is the propagation delay measured from the time the $\overline{\text{RESET}}$ pin is first asserted low to the time the SDA pin is asserted high, signaling a stop condition. It must be a minimum of t_{WL}.

6.9 Typical Characteristics

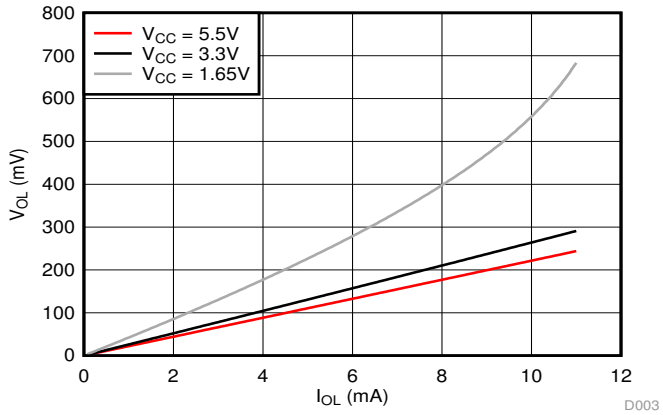


Figure 1. SDA Output Low Voltage (V_{OL}) vs Load Current (I_{OL}) at Three V_{CC} Levels

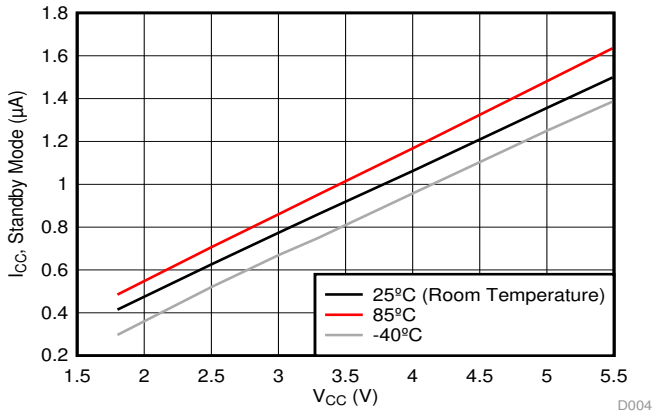


Figure 2. Standby Current (I_{CC}) vs Supply Voltage (V_{CC}) at Three Temperature Points

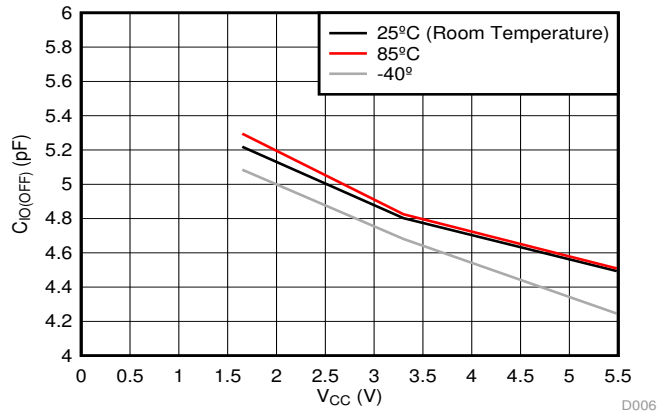


Figure 3. Slave channel (SCn/SDn) capacitance ($C_{io(OFF)}$) vs. Supply Voltage (V_{CC}) at Three Temperature Points

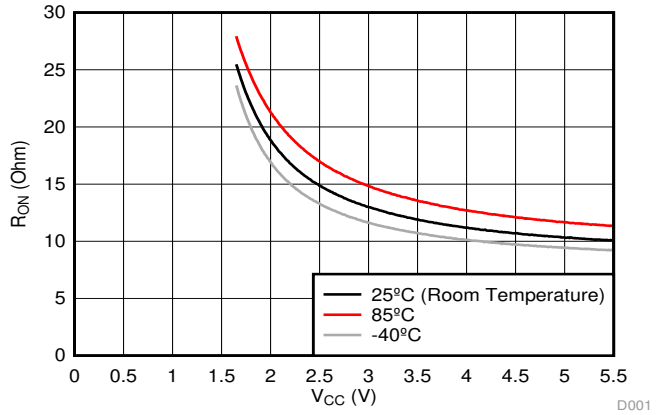
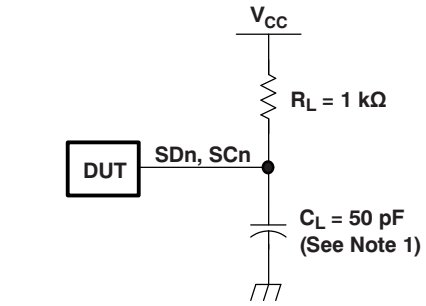


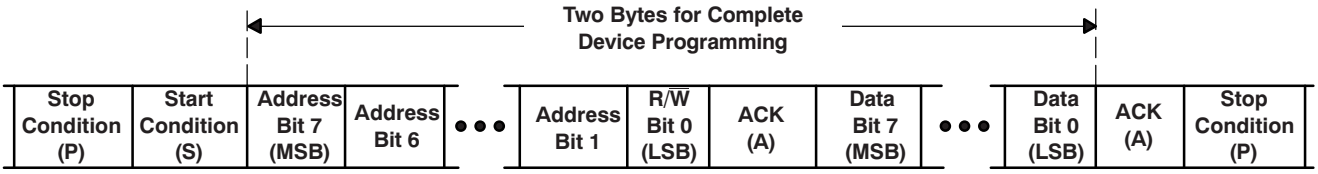
Figure 4. ON-Resistance (R_{ON}) vs Supply Voltage (V_{CC}) at Three Temperatures

7 Parameter Measurement Information

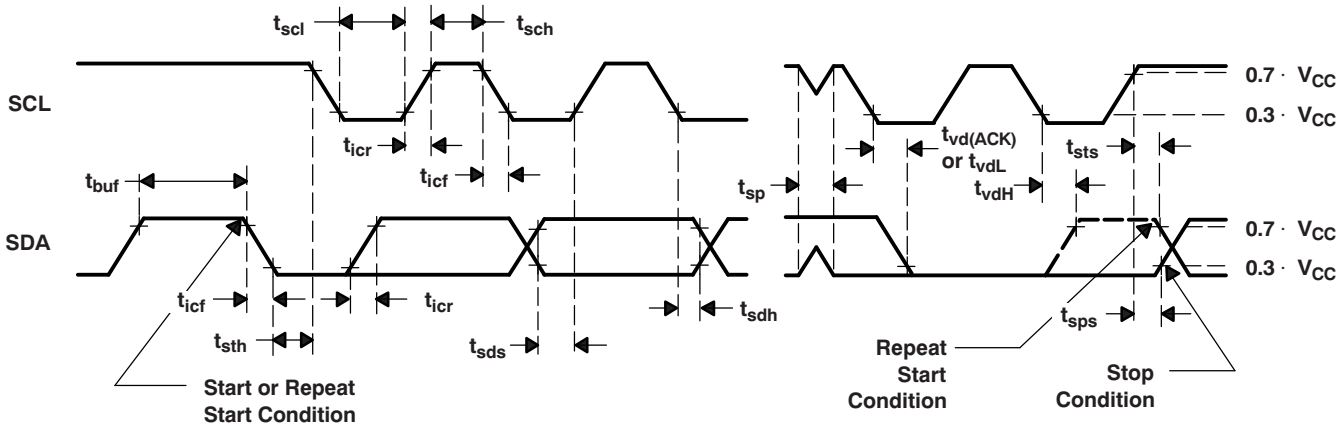


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I²C PORT LOAD CONFIGURATION



BYTE	DESCRIPTION
1	I^2C address + R/$\overline{W}$
2	Control register data



VOLTAGE WAVEFORMS

A. C_L includes probe and jig capacitance.

B. All input pulses are supplied by generators having the following characteristics: PRR ≤ 10 MHz, Z_O = 50 Ω, t_r/t_f ≤ 30 ns.

C. The outputs are measured one at a time, with one transition per measurement.

Figure 5. I^2C Interface Load Circuit, Byte Descriptions, and Voltage Waveforms

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Parameter Measurement Information (continued)

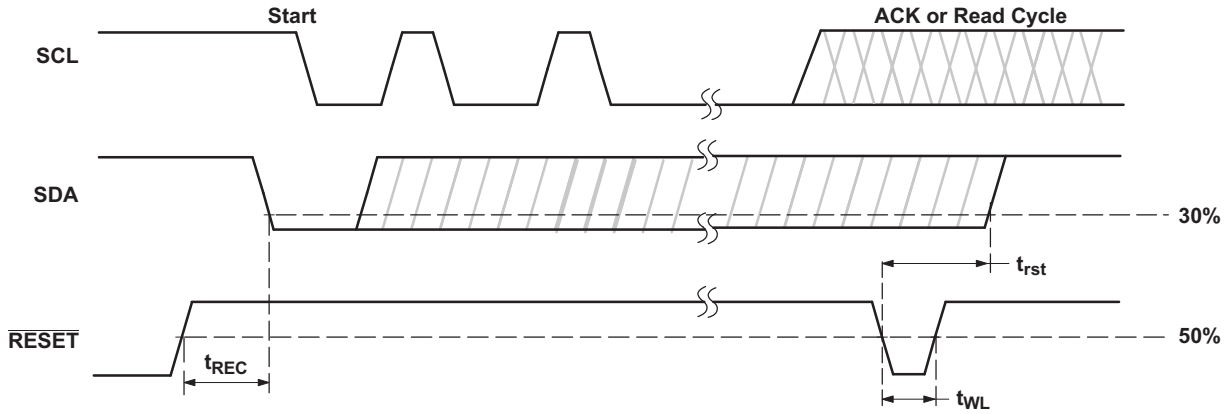


Figure 6. Reset Timing

8 Detailed Description

8.1 Overview

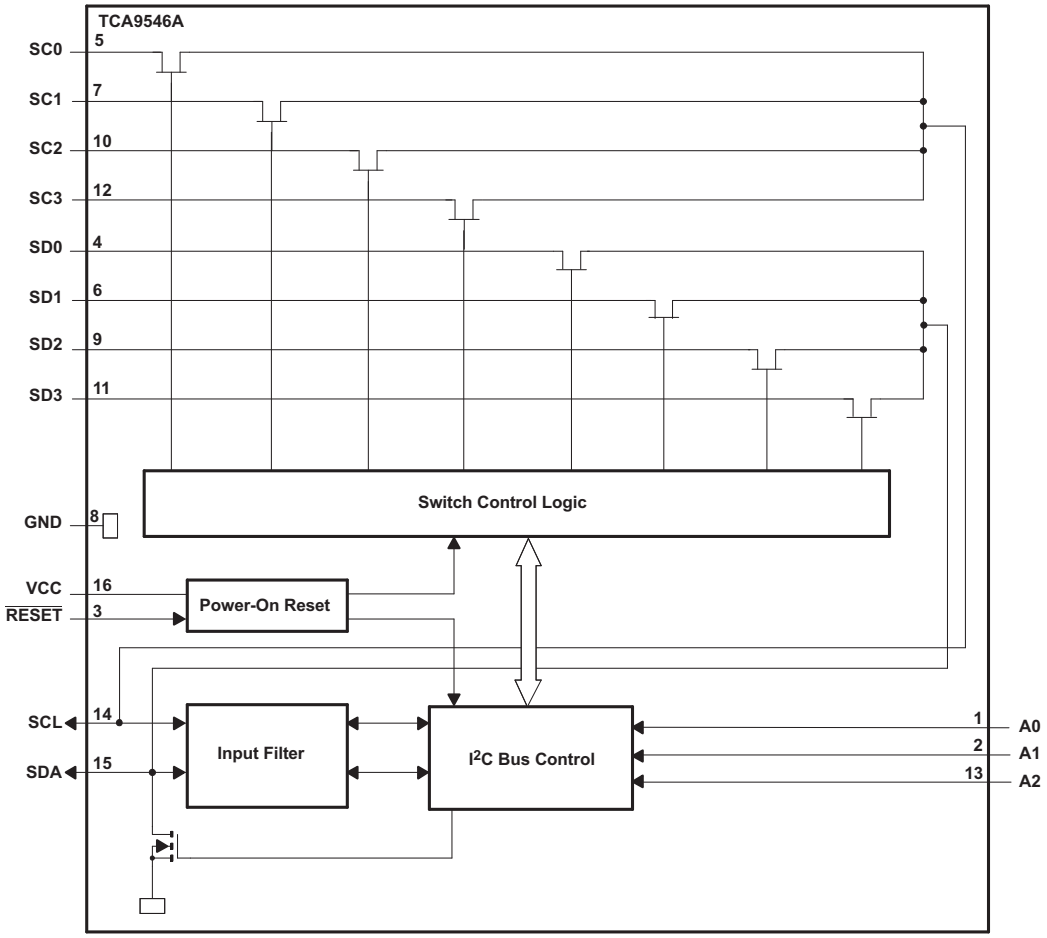
The TCA9546A is a 4-channel, bidirectional translating I^2C switch. The master SCL/SDA signal pair is directed to four channels of slave devices, SC0/SD0-SC3/SD3. Any individual downstream channel can be selected as well as any combination of the four channels.

The device offers an active-low $\overline{\text{RESET}}$ input which resets the state machine and allows the TCA9546A to recover should one of the downstream I^2C buses get stuck in a low state. The state machine of the device can also be reset by cycling the power supply, V_{CC}, also known as a power-on reset (POR). Both the $\overline{\text{RESET}}$ function and a POR will cause all channels to be deselected.

The connections of the I^2C data path are controlled by the same I^2C master device that is switched to communicate with multiple I^2C slaves. After the successful acknowledgment of the slave address (hardware selectable by A0 and A1 pins), a single 8-bit control register is written to or read from to determine the selected channels.

The TCA9546A may also be used for voltage translation, allowing the use of different bus voltages on each SCn/SDn pair such that 1.8-V, 2.5-V, or 3.3-V parts can communicate with 5-V parts. This is achieved by using external pull-up resistors to pull the bus up to the desired voltage for the master and each slave channel.

8.2 Functional Block Diagram



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8.3 Feature Description

The TCA9546A is a 4-channel, bidirectional translating switch for I^2C buses that supports Standard-Mode (100 kHz) and Fast-Mode (400 kHz) operation. The TCA9546A features I^2C control using a single 8-bit control register in which the four least significant bits control the enabling and disabling of the 4 switch channels of I^2C data flow. Depending on the application, voltage translation of the I^2C bus can also be achieved using the TCA9546A to allow 1.8-V, 2.5-V, or 3.3-V parts to communicate with 5-V parts. Additionally, in the event that communication on the I^2C bus enters a fault state, the TCA9546A can be reset to resume normal operation using the $\overline{\text{RESET}}$ pin feature or by a power-on reset which results from cycling power to the device.

8.4 Device Functional Modes

8.4.1 $\overline{\text{RESET}}$ Input

The $\overline{\text{RESET}}$ input is an active-low signal that may be used to recover from a bus-fault condition. When this signal is asserted low for a minimum of t_{WL}, the TCA9546A resets its registers and I^2C state machine and deselects all channels. The $\overline{\text{RESET}}$ input must be connected to V_{CC} through a pull-up resistor.

8.4.2 Power-On Reset

When power is applied to V_{CC}, an internal power-on reset holds the TCA9546A in a reset condition until V_{CC} has reached V_{POR}. At this point, the reset condition is released, and the TCA9546A registers and I^2C state machine are initialized to their default states, all zeroes, causing all the channels to be deselected. Thereafter, V_{CC} must be lowered below V_{POR} to reset the device.

8.5 Programming

8.5.1 I^2C Interface

The I^2C bus is for two-way two-line communication between different ICs or modules. The two lines are a serial data line (SDA) and a serial clock line (SCL). Both lines must be connected to a positive supply via a pullup resistor when connected to the output stages of a device. Data transfer can be initiated only when the bus is not busy.

One data bit is transferred during each clock pulse. The data on the SDA line must remain stable during the high period of the clock pulse, as changes in the data line at this time are interpreted as control signals (see Figure 7).

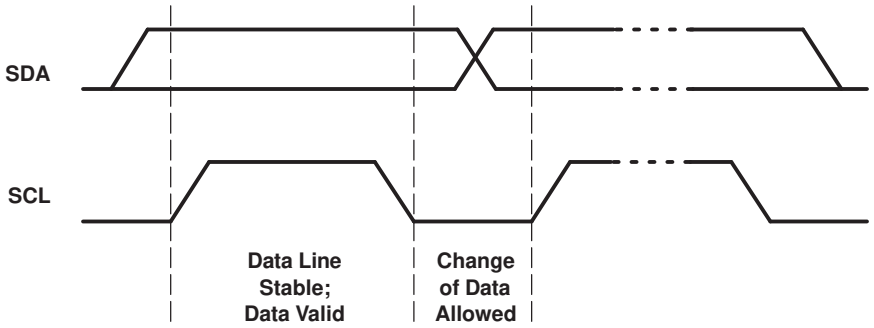


Figure 7. Bit Transfer

Both data and clock lines remain high when the bus is not busy. A high-to-low transition of the data line while the clock is high is defined as the start condition (S). A low-to-high transition of the data line while the clock is high is defined as the stop condition (P) (see Figure 8).

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Programming (continued)

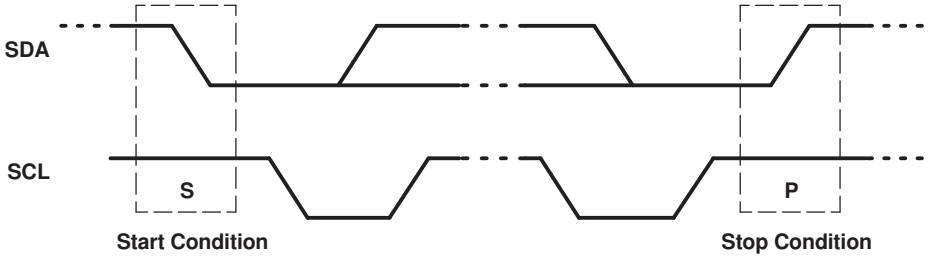


Figure 8. Definition of Start and Stop Conditions

A device generating a message is a transmitter; a device receiving is the receiver. The device that controls the message is the master, and the devices that are controlled by the master are the slaves (see Figure 9).

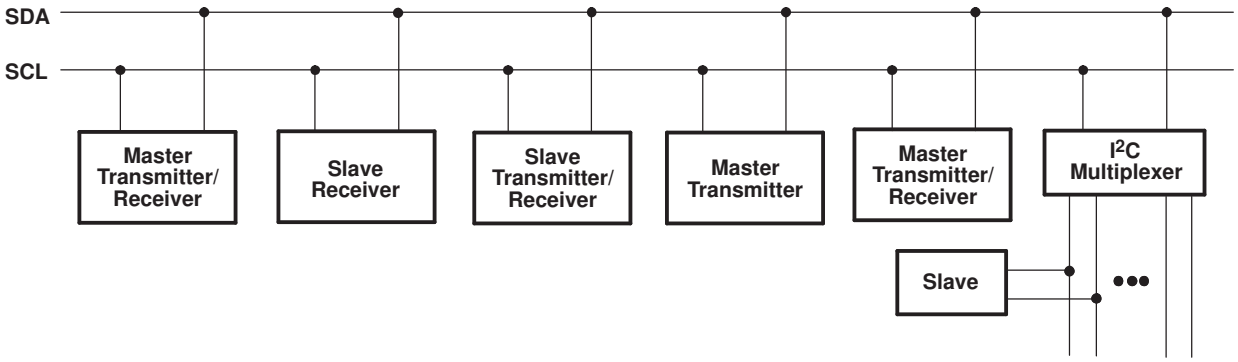


Figure 9. System Configuration

The number of data bytes transferred between the start and the stop conditions from transmitter to receiver is not limited. Each byte of eight bits is followed by one acknowledge (ACK) bit. The transmitter must release the SDA line before the receiver can send an ACK bit.

When a slave receiver is addressed, it must generate an ACK after the reception of each byte. Also, a master must generate an ACK after the reception of each byte that has been clocked out of the slave transmitter. The device that acknowledges must pull down the SDA line during the ACK clock pulse so that the SDA line is stable low during the high pulse of the ACK-related clock period (see Figure 10). Setup and hold times must be taken into account.

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Programming (continued)

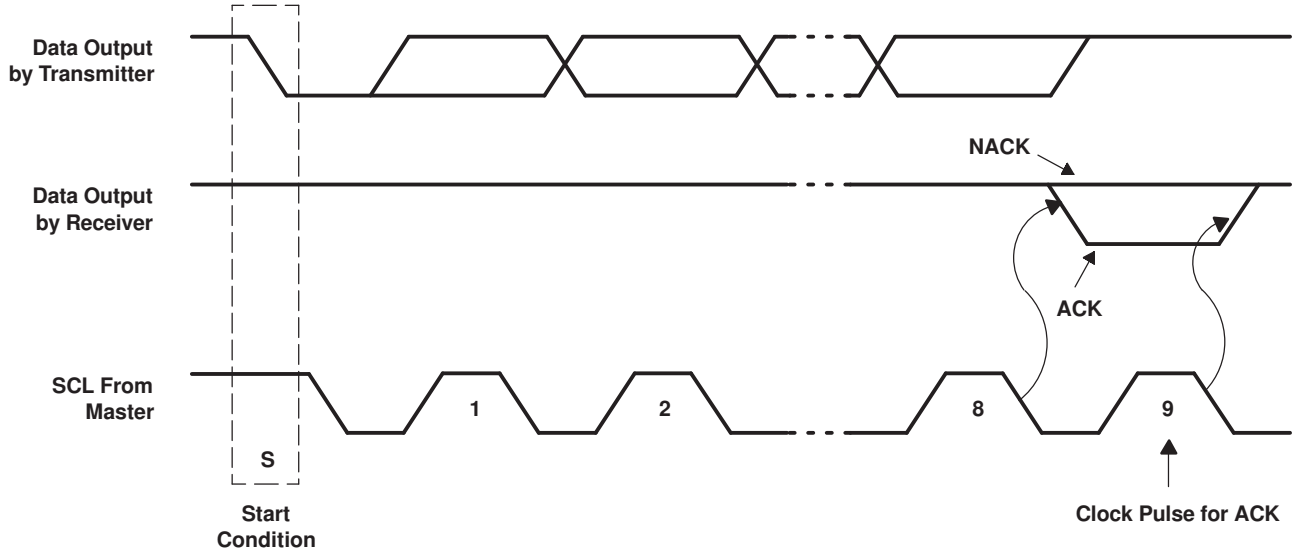


Figure 10. Acknowledgment on the I²C Bus

Data is transmitted to the TCA9546A control register using the write mode shown in Figure 11.

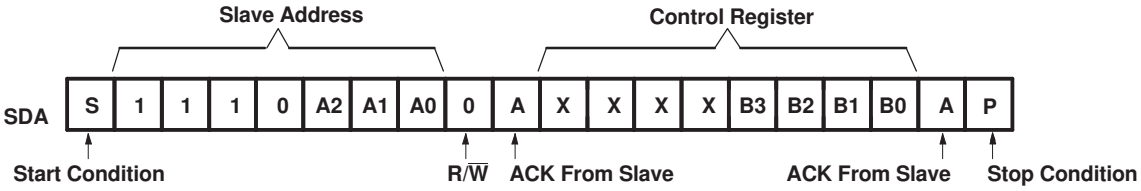


Figure 11. Write Control Register

Data is read from the TCA9546A control register using the read mode shown in Figure 12.

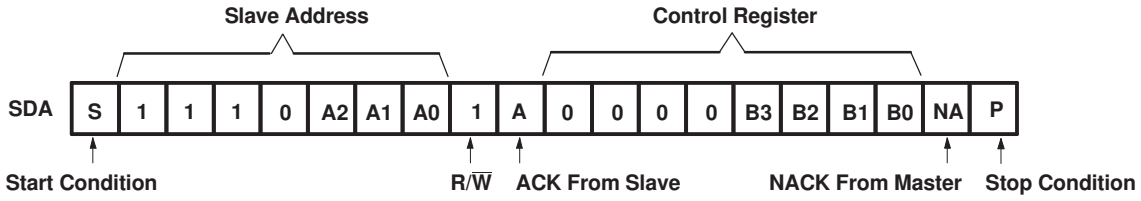


Figure 12. Read Control Register

8.6 Control Register

8.6.1 Device Address

Following a start condition, the bus master must output the address of the slave it is accessing. The address of the TCA9546A is shown in Figure 13. To conserve power, no internal pullup resistors are incorporated on the hardware-selectable address pins, and they must be pulled high or low.

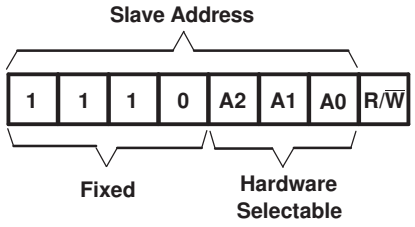


Figure 13. TCA9546A Address

The last bit of the slave address defines the operation to be performed. When set to a logic 1, a read is selected, while a logic 0 selects a write operation.

8.6.2 Control Register Description

Following the successful acknowledgment of the slave address, the bus master sends a byte to the TCA9546A, which is stored in the control register (see Figure 14). If multiple bytes are received by the TCA9546A, it will save the last byte received. This register can be written and read via the I^2C bus.

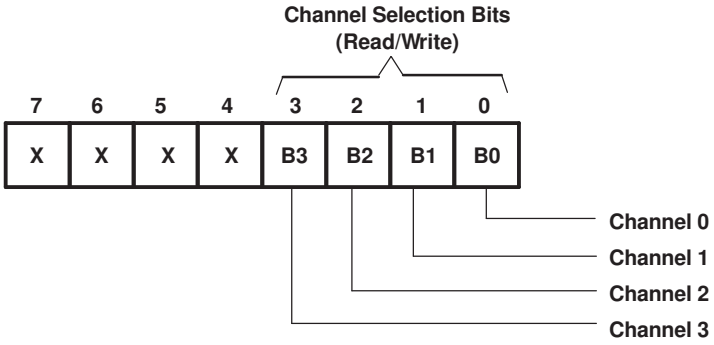


Figure 14. Control Register

8.6.3 Control Register Definition

One or several SCn/SDn downstream pairs, or channels, are selected by the contents of the control register (see Table 1). This register is written after the TCA9546A has been addressed. The four LSBs of the control byte are used to determine which channel or channels are to be selected. When a channel is selected, the channel becomes active after a stop condition has been placed on the I^2C bus. This ensures that all SCn/SDn lines are in a high state when the channel is made active, so that no false conditions are generated at the time of connection. A stop condition always must occur right after the acknowledge cycle.

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Control Register (continued)

Table 1. Control Register Write (Channel Selection), Control Register Read (Channel Status)(1)

B7	B6	B5	B4	B3	B2	B1	B0	COMMAND
X	X	X	X	X	X	X	0	Channel 0 disabled
							1	Channel 0 enabled
X	X	X	X	X	X	0	X	Channel 1 disabled
						1		Channel 1 enabled
X	X	X	X	X	0	X	X	Channel 2 disabled
					1			Channel 2 enabled
X	X	X	X	0	X	X	X	Channel 3 disabled
				1				Channel 3 enabled
0	0	0	0	0	0	0	0	No channel selected, power-up/reset default state

(1) Several channels can be enabled at the same time. For example, B3 =0, B2 = 1, B1 = 1, B0 = 0 means that channels 0 and 3 are disabled, and channels 1 and 2 are enabled. Care should be taken not to exceed the maximum bus capacity.

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9 Application and Implementation

9.1 Application Information

Applications of the TCA9546A contains an I^2C (or SMBus) master device and up to four I^2C slave devices. The downstream channels are ideally used to resolve I^2C slave address conflicts. For example, if four identical digital temperature sensors are needed in the application, one sensor can be connected at each channel: 0, 1, 2, and 3. When the temperature at a specific location needs to be read, the appropriate channel can be enabled and all other channels switched off, the data can be retrieved, and the I^2C master can move on and read the next channel.

In an application where the I^2C bus contain many additional slave devices that do not result in I^2C slave address conflicts, these slave devices can be connected to any desired channel to distribute the total bus capacitance across multiple channels. If multiple switches will be enabled simultaneously, additional design requirements must be considered (See Design Requirements and Detailed Design Procedure).

9.2 Typical Application

A typical application of the TCA9546A contains anywhere from 1 to 5 separate data pull-up voltages, V_{DPUX} , one for the master device (V_{DPUM}) and one for each of the selectable slave channels (V_{DPU0} – V_{DPU3}). In the event where the master device and all slave devices operate at the same voltage, then the pass voltage, $V_{pass} = V_{DPUX}$. Once the maximum V_{pass} is known, V_{cc} can be selected easily using Figure 16. In an application where voltage translation is necessary, additional design requirements must be considered (See Design Requirements).

Figure 15 shows an application in which the TCA9546A can be used.

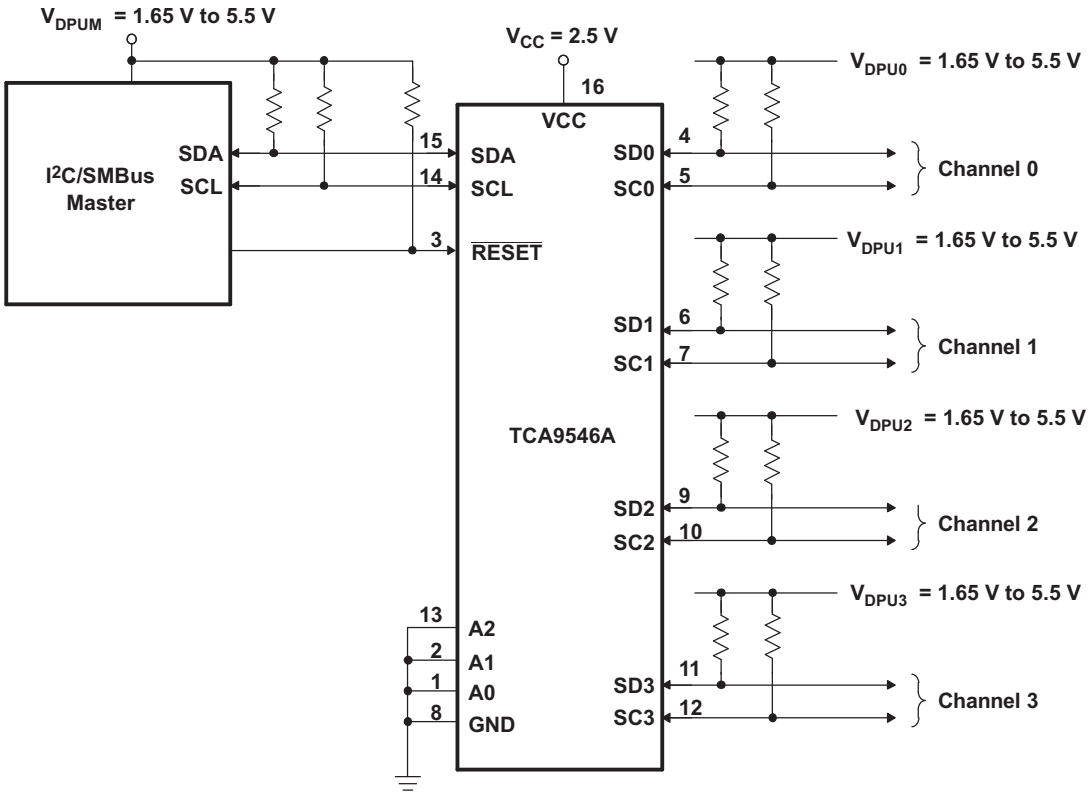


Figure 15. TCA9546A Typical Application Schematic

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Typical Application (continued)

9.2.1 Design Requirements

The A0, A1, and A2 pins are hardware selectable to control the slave address of the TCA9546A. These pins may be tied directly to GND or V_{CC} in the application.

If multiple slave channels will be activated simultaneously in the application, then the total I_{OL} from SCL/SDA to GND on the master side will be the sum of the currents through all pull-up resistors, R_p.

The pass-gate transistors of the TCA9546A are constructed such that the V_{CC} voltage can be used to limit the maximum voltage that is passed from one I^2C bus to another.

Figure 16 shows the voltage characteristics of the pass-gate transistors (note that the graph was generated using data specified in the Electrical Characteristics section of this data sheet). In order for the TCA9546A to act as a voltage translator, the V_{pass} voltage must be equal to or lower than the lowest bus voltage. For example, if the main bus is running at 5 V and the downstream buses are 3.3 V and 2.7 V, V_{pass} must be equal to or below 2.7 V to effectively clamp the downstream bus voltages. As shown in Figure 16, $V_{pass(max)}$ is 2.7 V when the TCA9546A supply voltage is 4 V or lower, so the TCA9546A supply voltage could be set to 3.3 V. Pull-up resistors then can be used to bring the bus voltages to their appropriate levels (see Figure 15).

9.2.2 Detailed Design Procedure

Once all the slaves are assigned to the appropriate slave channels and bus voltages are identified, the pull-up resistors, R_p, for each of the buses need to be selected appropriately. The minimum pull-up resistance is a function of V_{DPUX}, $V_{OL,(max)}$, and I_{OL}:

$$R_{p(min)} = \frac{V_{DPUX} - V_{OL(max)}}{I_{OL}} \quad (1)$$

The maximum pull-up resistance is a function of the maximum rise time, t_r (300 ns for fast-mode operation, f_{SCL} = 400 kHz) and bus capacitance, C_b:

$$R_{p(max)} = \frac{t_r}{0.8473 \times C_b} \quad (2)$$

The maximum bus capacitance for an I^2C bus must not exceed 400 pF for fast-mode operation. The bus capacitance can be approximated by adding the capacitance of the TCA9546A, $C_{io(OFF)}$, the capacitance of wires/connections/traces, and the capacitance of each individual slave on a given channel. If multiple channels will be activated simultaneously, each of the slaves on all channels will contribute to total bus capacitance.

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Typical Application (continued)

9.2.3 TCA9546A Application Curves

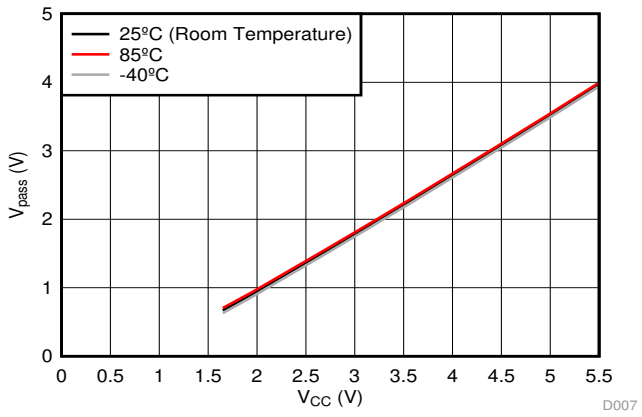
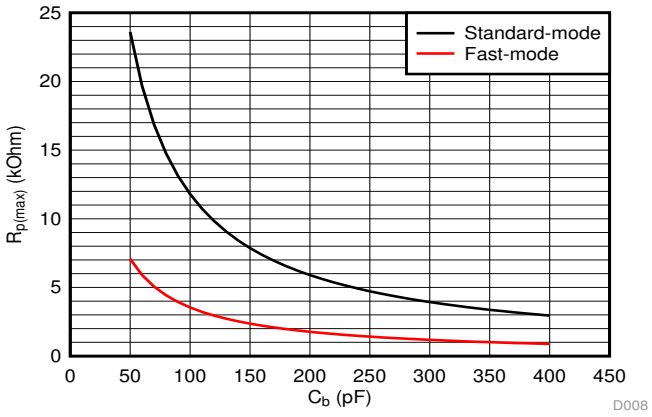


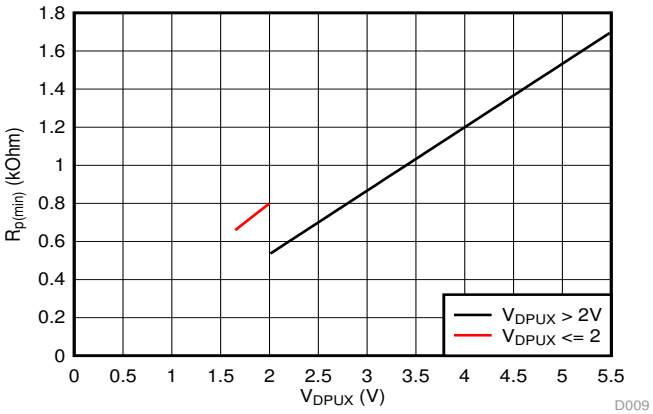
Figure 16. Pass-Gate Voltage (V_{pass}) vs Supply Voltage (V_{CC}) at Three Temperature Points



Standard-mode
(f_{SCL}= 100 kHz, t_r = 1 µs)

Fast-mode
(f_{SCL}= 400 kHz, t_r= 300 ns)

Figure 17. Maximum Pull-Up resistance ($R_{p(max)}$) vs Bus Capacitance (C_b)



V_{OL} = 0.2*V_{DPUX}, I_{OL} = 2 mA when V_{DPUX} ≤ 2 V
V_{OL} = 0.4 V, I_{OL} = 3 mA when V_{DPUX} > 2 V

Figure 18. Minimum Pull-Up Resistance ($R_{p(min)}$) vs Pull-Up Reference Voltage (V_{DPUX})

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10 Power Supply Recommendations

The operating power-supply voltage range of the TCA9546A is 1.65 V to 5.5 V applied at the VCC pin. When the TCA9546A is powered on for the first time or anytime the device needs to be reset by cycling the power supply, the power-on reset requirements must be followed to ensure the I^2C bus logic is initialized properly.

10.1 Power-On Reset Requirements

In the event of a glitch or data corruption, TCA9546A can be reset to its default conditions by using the power-on reset feature. Power-on reset requires that the device go through a power cycle to be completely reset. This reset also happens when the device is powered on for the first time in an application.

A power-on reset is shown in Figure 19.

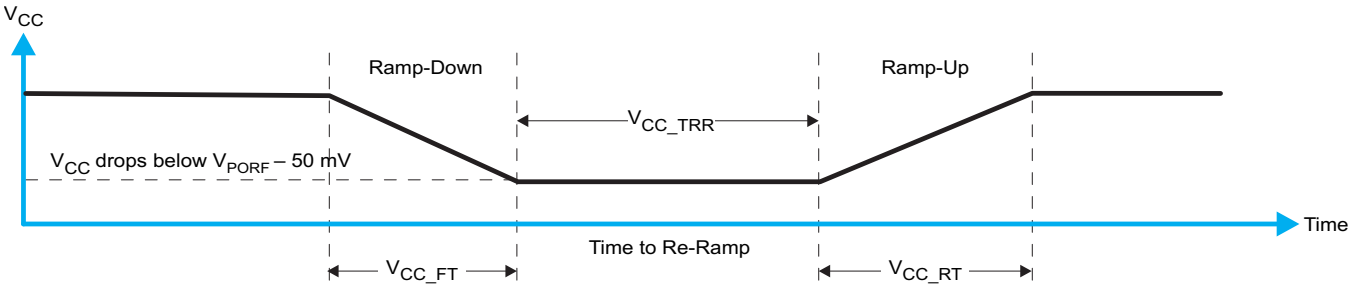


Figure 19. V_{CC} is Lowered Below the POR Threshold, Then Ramped Back Up to V_{CC}

Table 2 specifies the performance of the power-on reset feature for TCA9546A for both types of power-on reset.

Table 2. Recommended Supply Sequencing And Ramp Rates(1)

PARAMETER			MIN	TYP	MAX	UNIT
V_{CC_FT}	Fall time	See Figure 19	1		100	ms
V_{CC_RT}	Rise time	See Figure 19	0.1		100	ms
V_{CC_TRR}	Time to re-ramp (when V_{CC} drops below $V_{PORF(min)}$ – 50 mV or when V_{CC} drops to GND)	See Figure 19	40			µs
V_{CC_GH}	Level that V_{CC} can glitch down to, but not cause a functional disruption when V_{CC_GW} = 1 µs	See Figure 20			1.2	V
V_{CC_GW}	Glitch width that will not cause a functional disruption when V_{CC_GH} = 0.5 × V_{CC}	See Figure 20			10	µs
V_{PORF}	Voltage trip point of POR on falling V_{CC}	See Figure 21	0.8		1.25	V
V_{PORR}	Voltage trip point of POR on rising V_{CC}	See Figure 21	1.05		1.5	V

(1) All supply sequencing and ramp rate values are measured at T_A = 25°C

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Glitches in the power supply can also affect the power-on reset performance of this device. The glitch width (V_{CC_GW}) and height (V_{CC_GH}) are dependent on each other. The bypass capacitance, source impedance, and device impedance are factors that affect power-on reset performance. Figure 20 and Table 2 provide more information on how to measure these specifications.

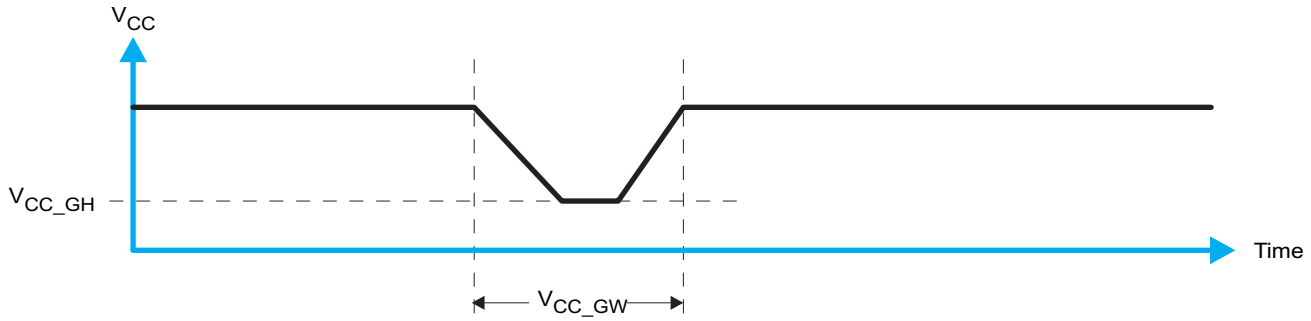


Figure 20. Glitch Width and Glitch Height

V_{POR} is critical to the power-on reset. V_{POR} is the voltage level at which the reset condition is released and all the registers and the I^2C/SMBus state machine are initialized to their default states. The value of V_{POR} differs based on the V_{CC} being lowered to or from 0. Figure 21 and Table 2 provide more details on this specification.

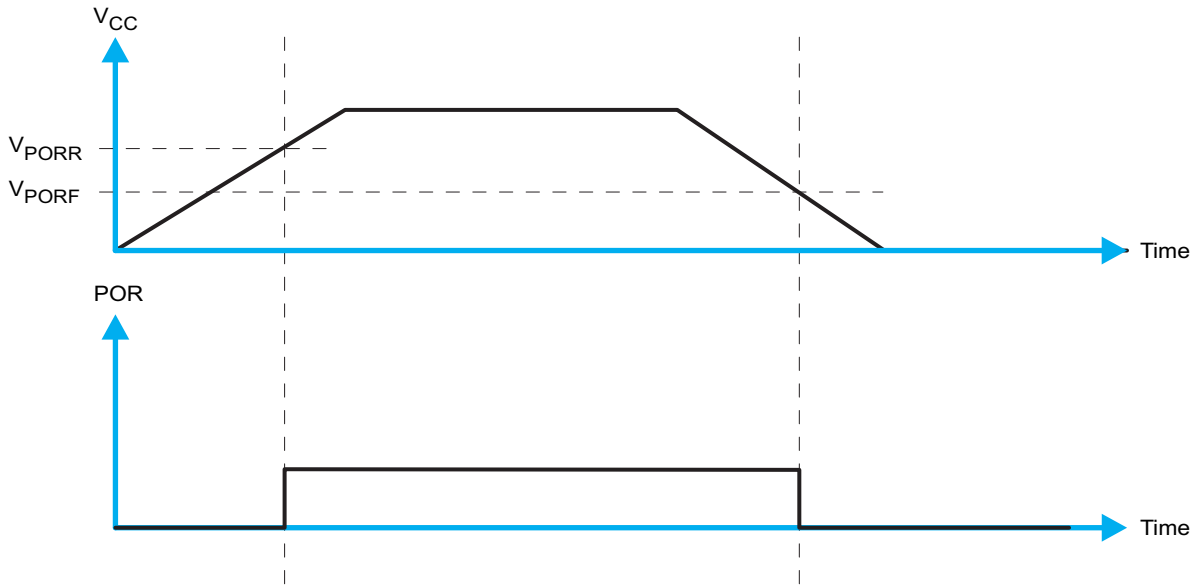


Figure 21. V_{POR}

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11 Layout

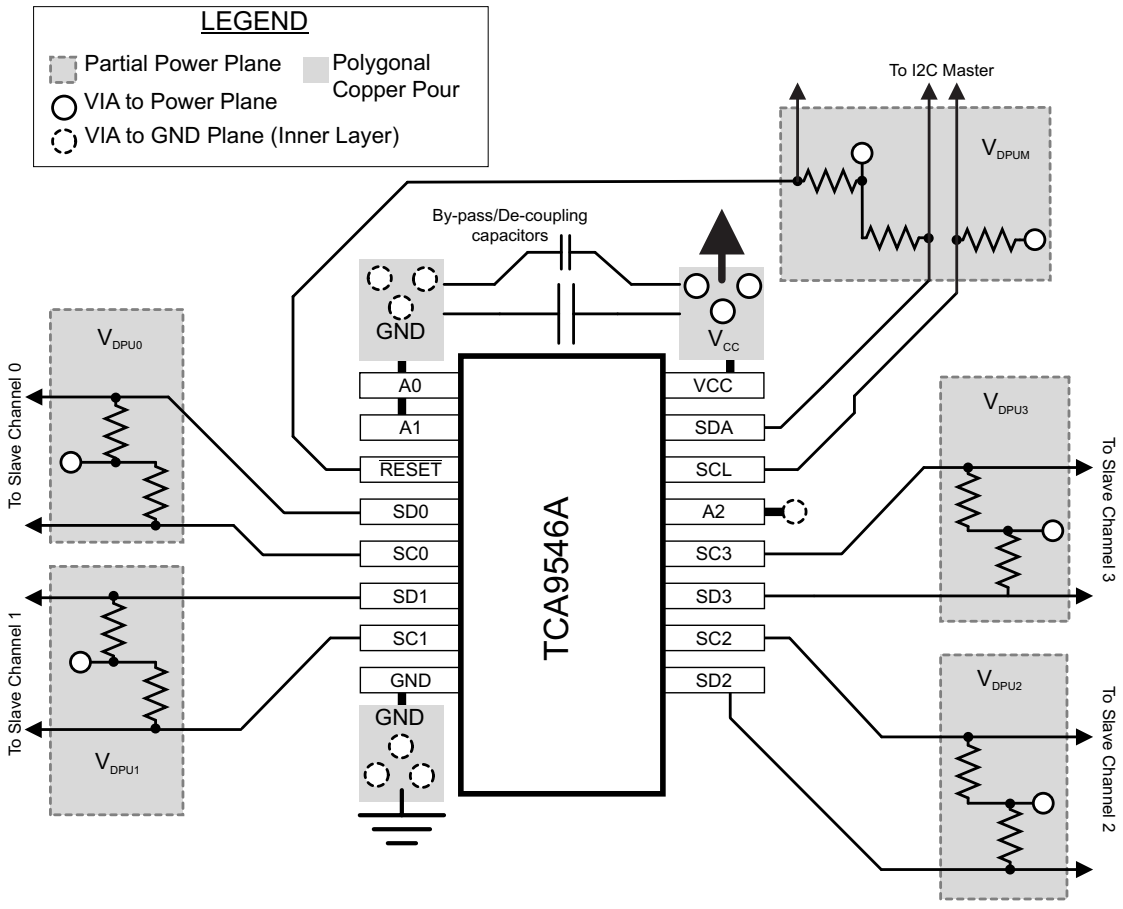
11.1 Layout Guidelines

For PCB layout of the TCA9546A, common PCB layout practices should be followed but additional concerns related to high-speed data transfer such as matched impedances and differential pairs are not a concern for I^2C signal speeds. It is common to have a dedicated ground plane on an inner layer of the board and pins that are connected to ground should have a low-impedance path to the ground plane in the form of wide polygon pours and multiple vias. By-pass and de-coupling capacitors are commonly used to control the voltage on the VCC pin, using a larger capacitor to provide additional power in the event of a short power supply glitch and a smaller capacitor to filter out high-frequency ripple.

In an application where voltage translation is not required, all V_{DPUX} voltages and V_{CC} could be at the same potential and a single copper plane could connect all of pull-up resistors to the appropriate reference voltage. In an application where voltage translation is required, V_{DPUM}, V_{DPU0}, V_{DPU1}, V_{DPU2}, and V_{DPU3} may all be on the same layer of the board with split planes to isolate different voltage potentials.

To reduce the total I^2C bus capacitance added by PCB parasitics, data lines (SCn and SDn) should be a short as possible and the widths of the traces should also be minimized (e.g. 5-10 mils depending on copper weight).

11.2 Layout Example



12 Device and Documentation Support

12.1 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on ti.com. In the upper right corner, click on *Alert me* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

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12.4 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

12.5 Glossary

SLYZ022 — *TI Glossary*.

This glossary lists and explains terms, acronyms, and definitions.

13 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation

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PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package | Pins	Package qty | Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
TCA9546ADR	Active	Production	SOIC (D) | 16	2500 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	TCA9546A
TCA9546ADR.B	Active	Production	SOIC (D) | 16	2500 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	TCA9546A
TCA9546APWR	Active	Production	TSSOP (PW) | 16	2000 | LARGE T&R	Yes	NIPDAU | SN	Level-1-260C-UNLIM	-40 to 85	PW546A
TCA9546APWR.B	Active	Production	TSSOP (PW) | 16	2000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	PW546A
TCA9546APWRG4	Active	Production	TSSOP (PW) | 16	2000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	PW546A
TCA9546APWRG4.B	Active	Production	TSSOP (PW) | 16	2000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	PW546A

(1) **Status:** For more details on status, see our product life cycle.

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

(3) **RoHS values:** Yes, No, RoHS Exempt. See the TI RoHS Statement for additional information and value definition.

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

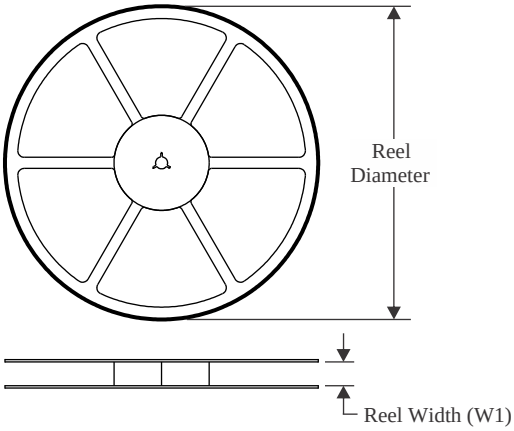
Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

Important Information and Disclaimer: The information provided on this page represents TI's knowledge and belief as of the date that it is provided. TI bases its knowledge and belief on information provided by third parties, and makes no representation or warranty as to the accuracy of such information. Efforts are underway to better integrate information from third parties. TI has taken and continues to take reasonable steps to provide representative and accurate information but may not have conducted destructive testing or chemical analysis on incoming materials and chemicals. TI and TI suppliers consider certain information to be proprietary, and thus CAS numbers and other limited information may not be available for release.

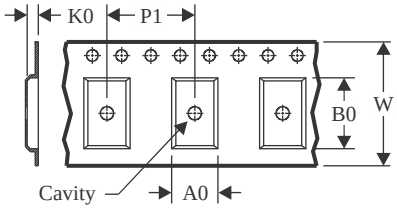
In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

TAPE AND REEL INFORMATION

REEL DIMENSIONS

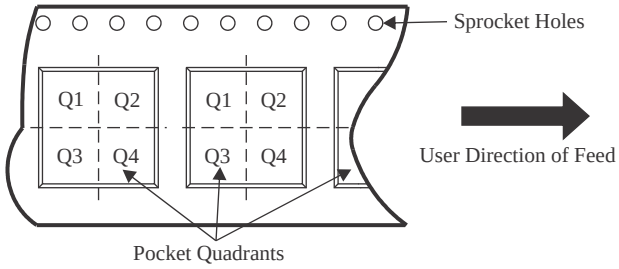


TAPE DIMENSIONS



A0	Dimension designed to accommodate the component width
B0	Dimension designed to accommodate the component length
K0	Dimension designed to accommodate the component thickness
W	Overall width of the carrier tape
P1	Pitch between successive cavity centers

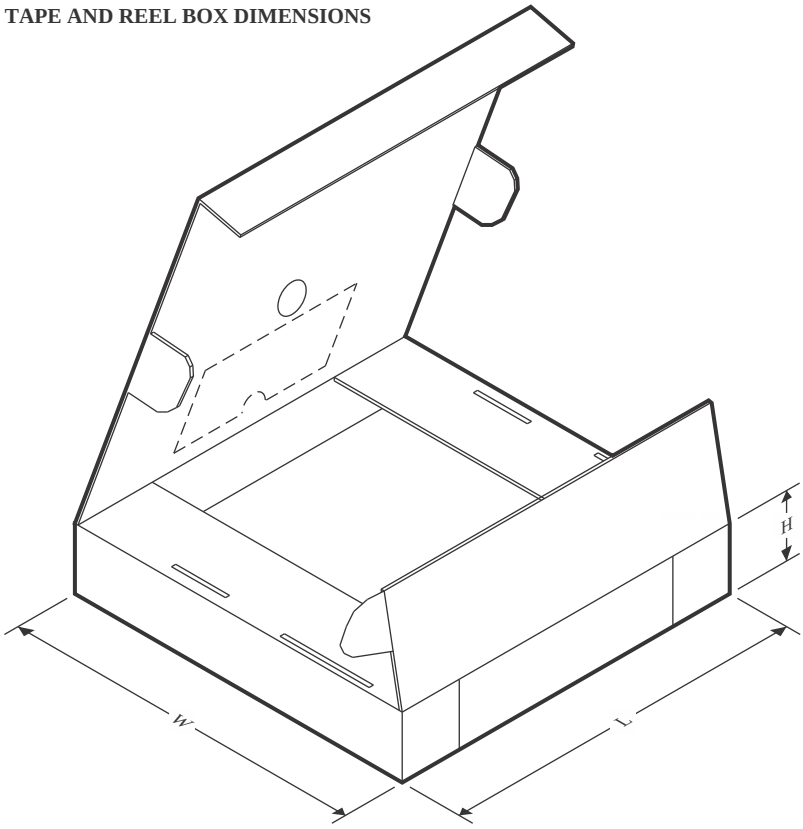
QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
TCA9546ADR	SOIC	D	16	2500	330.0	16.4	6.5	10.3	2.1	8.0	16.0	Q1
TCA9546APWR	TSSOP	PW	16	2000	330.0	12.4	6.9	5.6	1.6	8.0	12.0	Q1
TCA9546APWR	TSSOP	PW	16	2000	330.0	12.4	6.9	5.3	1.6	8.0	12.0	Q1
TCA9546APWR	TSSOP	PW	16	2000	330.0	12.4	6.9	5.6	1.6	8.0	12.0	Q1
TCA9546APWRG4	TSSOP	PW	16	2000	330.0	12.4	6.9	5.6	1.6	8.0	12.0	Q1

TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

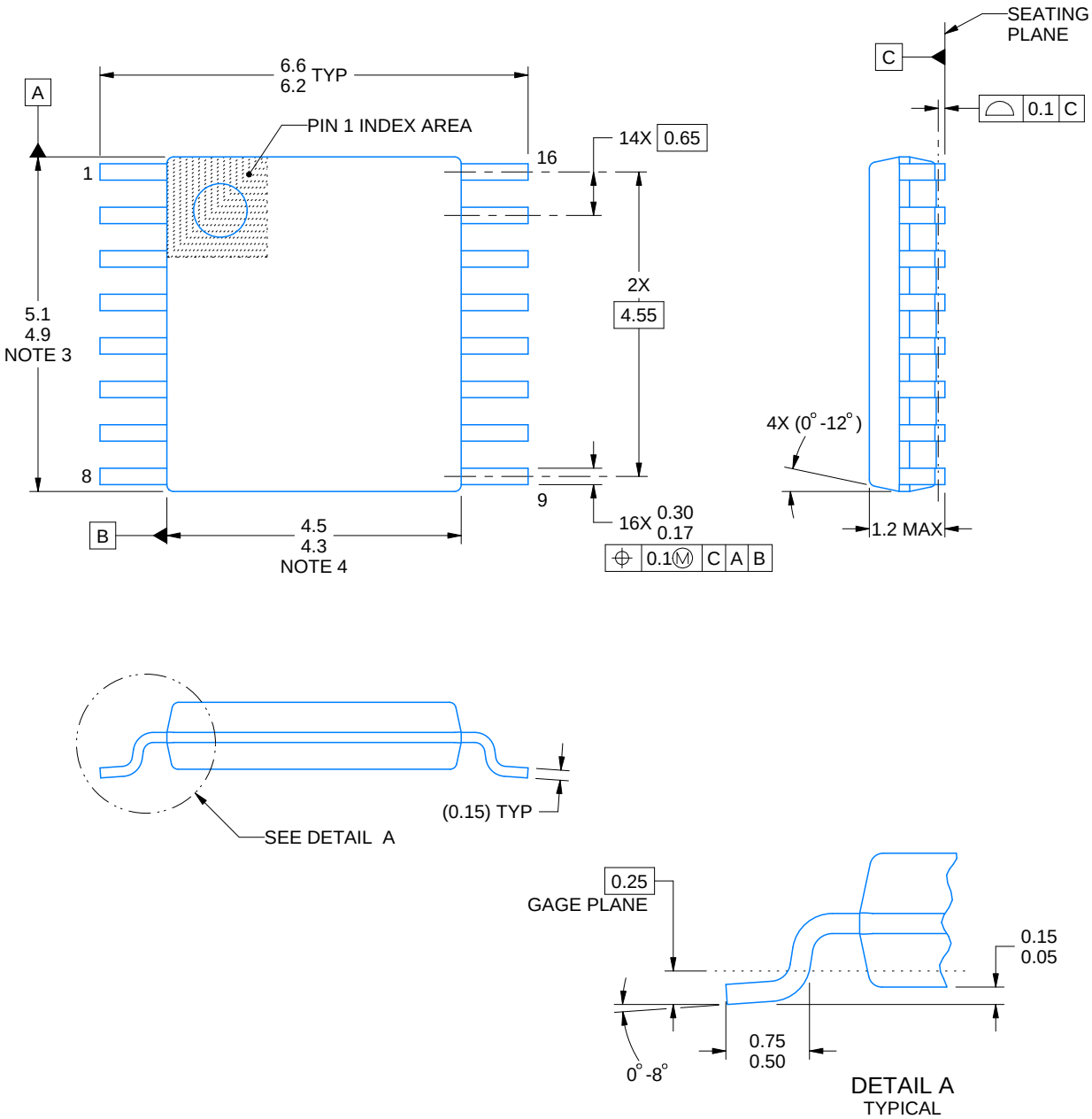
Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
TCA9546ADR	SOIC	D	16	2500	353.0	353.0	32.0
TCA9546APWR	TSSOP	PW	16	2000	353.0	353.0	32.0
TCA9546APWR	TSSOP	PW	16	2000	367.0	367.0	35.0
TCA9546APWR	TSSOP	PW	16	2000	356.0	356.0	35.0
TCA9546APWRG4	TSSOP	PW	16	2000	353.0	353.0	32.0

PACKAGE OUTLINE

PW0016A

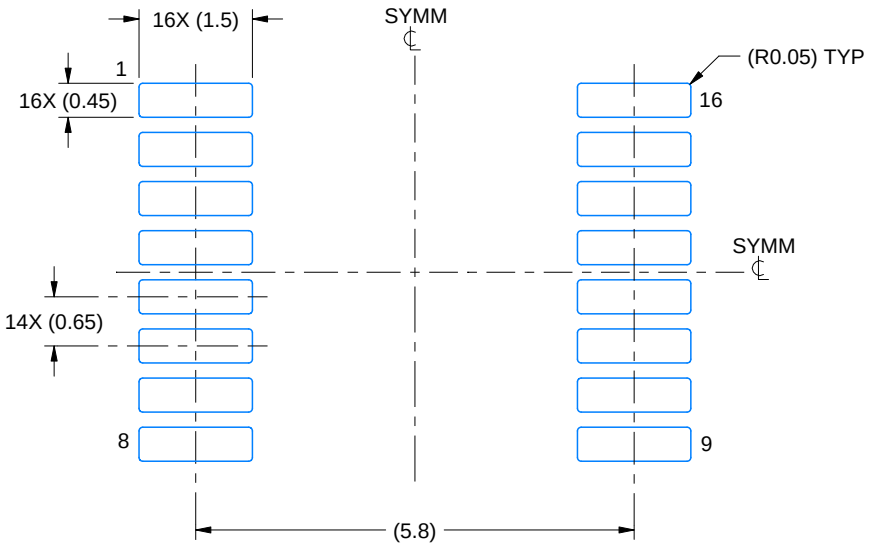
TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE

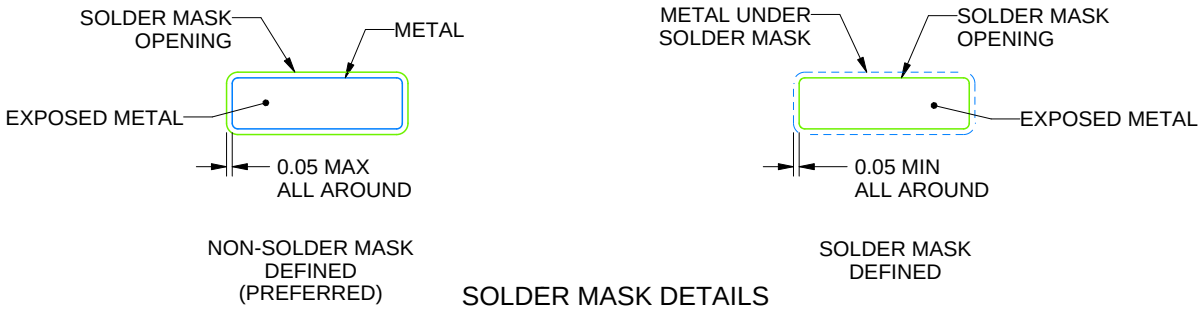


NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm per side.
5. Reference JEDEC registration MO-153.



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE: 10X

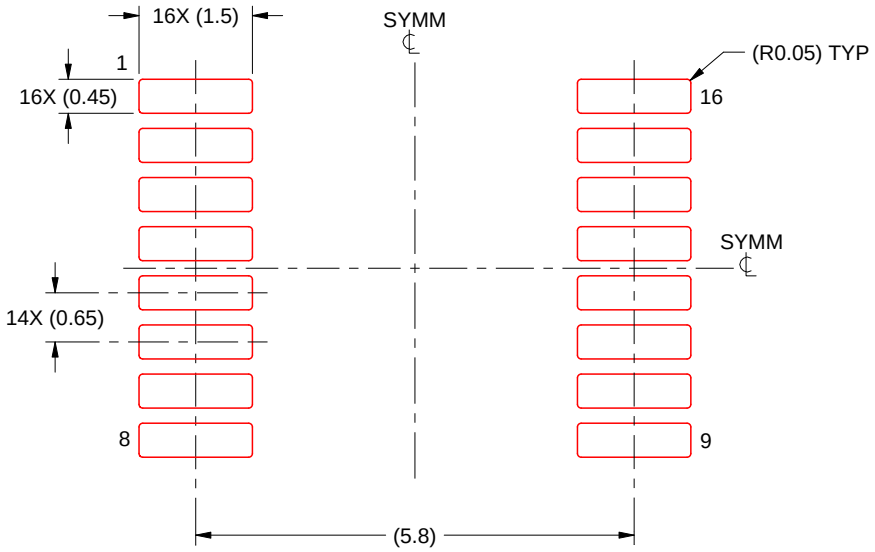


SOLDER MASK DETAILS

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NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.



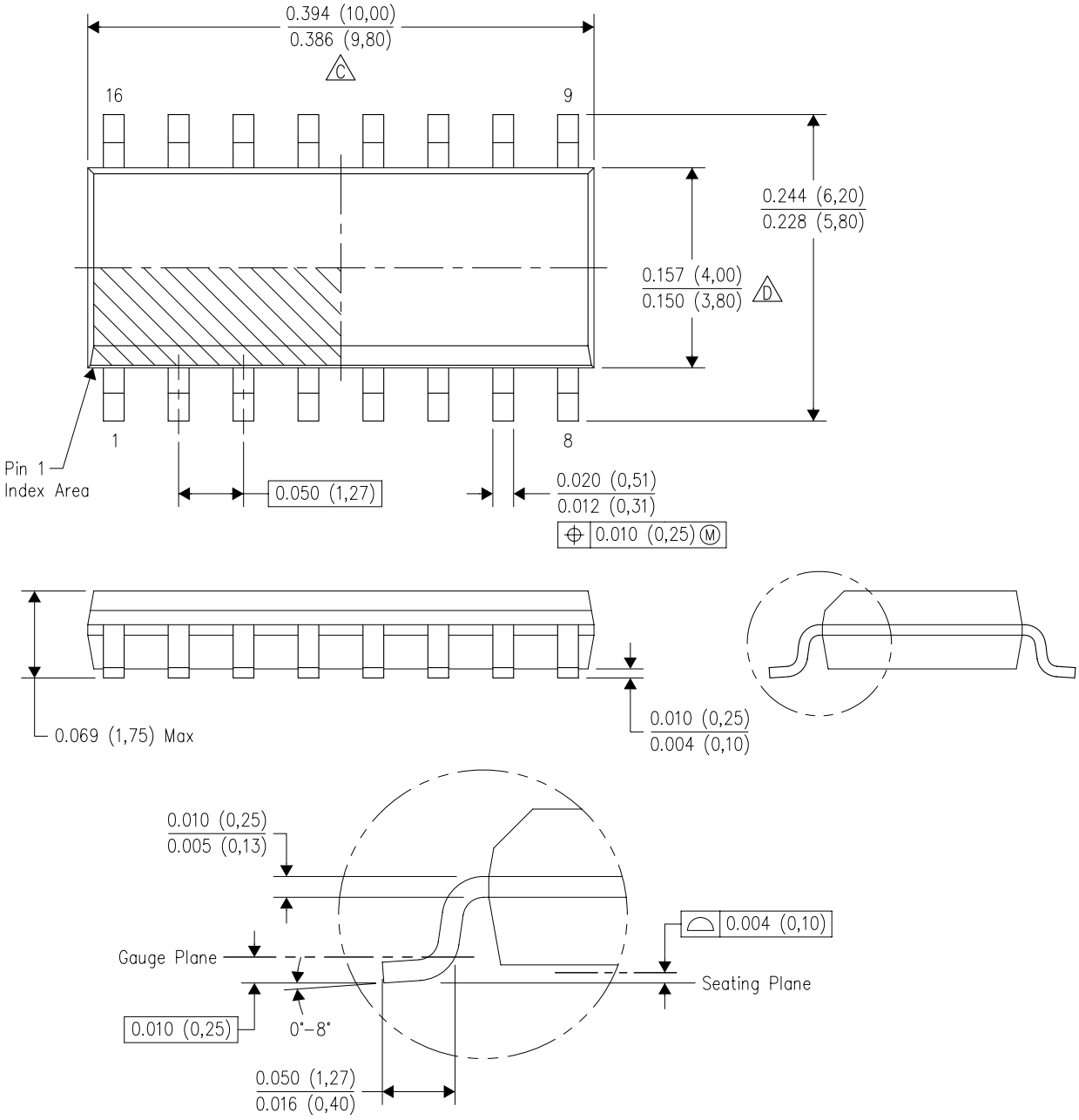
SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE: 10X

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NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

D (R–PDSO–G16) PLASTIC SMALL OUTLINE



NOTES:
A. All linear dimensions are in inches (millimeters).
B. This drawing is subject to change without notice.
C. Body length does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.006 (0,15) each side.
D. Body width does not include interlead flash. Interlead flash shall not exceed 0.017 (0,43) each side.
E. Reference JEDEC MS–012 variation AC.

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